This IV Exam will open at XX on XX and close at XX on XX.

There is a 90 minute time limit for this exam. You are only allowed one attempt at it.

In section A, there are 15 multiple choice questions, each with three answer options. Each question is worth up to two marks.

Each question in section A has one (and only one) correct answer option. If you think that more than one answer is correct, please select the answer you think is most likely or most important. Each correct answer earns you two marks; each incorrect answer incurs a one mark penalty.

Section B presents three further 10-mark questions, requiring more traditional written responses.

The total will be scaled to account for 70% of your total mark for IV.

SECTION A

1. A programmer records their visualisation program's performance with a number of random samples from an input data set, noting (for each test) the sample size, the metric used, the run time, and a unique identifier for the text, e.g. (10000, Manhattan, 23.3, 1) for the first test. How best to describe the recorded data?
    a) data could be treated as a table, with each row being a tuple of three items: two sequential quantitative items, and one categorical item
    b) The data could be treated as a table, with each row being a tuple of four items: three sequential quantitative items, and one categorical item
    c) The data could be treated as a table, with each row being a tuple of four items: two sequential quantitative items, and two categorical items

2. A programmer runs ten tests of their visualisation system, recording run time and data set size for each test (along with an identifier for each test). If they were to make a chart of the run time for each test, to look for patterns and outliers, which of the following would be best?
    a) A pie chart, with ten segments, one segment per test. Each segment's area matches the proportion of the total of all ten run times.
    b) A bar chart with identifiers spaced evenly along the x-axis, and run time as the height of the bar above each identifier
    c) Aline chart connecting 10 dots, with the dots spaced evenly along the x-axis, and run time shown as the y coordinate of each dot

3. A programmer has collected data on run time and data set size from 100 tests of their program. They now plan to make a scatterplot with data set size as the x-axis, and run time as the y-axis. Which of the following best describe the design choice for the mark on the scatterplot, made for each test?
    a) Each mark would be a small dot, and the identifier would be encoded as the colour hue of that dot
    b) It is not a good idea to encode the identifier for each test into a visual variable for each mark
    c) Each mark would be a small dot, and the identifier would be encoded as the colour value of that dot

4. When depicting bivariate data, where one variable is quantitative and the other is categorical:
    a) a scatterplot might be a good depiction, but no trend lines should be drawn

b) a line chart will not be a good depiction, unless the category order is fairly randomised
c) a bar chart could be a good depiction, if the zero bar height matches the minimum quantitative value

5. Which of these statements is true?
    a) Quantitative data that is univariate can be represented in a scatterplot.
    b) Quantitative data that is bivariate can be represented in a pie chart.
    c) Quantitative data that is trivariate can be represented using a spring model.

6. A supermarket is working with two data sets, A and B. A consists of the names of the 10 food products most often bought by _each_ customer. B consists of the number of times each customer has bought each of the 10 products most frequently bought in the supermarket, based on data from _all_ customers. Which of the following best describes the data sets, using Munzner's classification?
    a) Two tables
    b) A table, and a multidimensional table
    c) Two multidimensional tables

7. Imagine you are the person responsible for formative evaluation of an application in a company, doing interviews with test participants (sometimes multiple interviews with the same participant), and managing a database to store all the qualitative data from that work. Which of the following database attributes would be most appropriate and useful for your work?
    a) ParticipantID, InterviewText, ProofOfWorthMeasure, IterationNumber
    b) ParticipantID, InterviewText, ProofOfWorthMeasure, InterviewNumber
    c) ParticipantID, InterviewText, IterationNumber, InterviewNumber

8. Imagine a data set of dates and locations of theatre shows. Selecting a rectangular area on a map of the data is used to create a declarative query. If the same query is then used in a linked bar chart, with one bar per month
    a) all the selected items will be in one bar
    b) the bars with the selected items will be next to each other
    c) the selected items could be in any combination of bars

9. How many scatterplots are in a SPLOM, used to present 9-dimensional data?
    a) 72
    b) 45
    c) 81

10. Imagine you are designing a system that some scientists can use to analyse some complex data. You will be using a dimensional reduction algorithm. The scientists say that fine-grained local detail is important, but also that the presence of high-level clusters (and accurate relationships between these clusters) is also important. Which of these algorithms would not be a good choice for this?
    a) Spring model
    b) tSNE
    c) UMAP

11. Imagine you are using a basic spring model algorithm, to analyse some complex data with 100 quantitative dimensions. The distance metric is simple geometric distance in the high-dimensional space. You first lay out 5000 objects, using just 50 dimensions. If you then plan to lay out 10000

objects, using the same 50 dimensions, roughly how long will each iteration of the spring model take?

a) 2 times longer than before
b) 16 times longer than before
c) 4 times longer than before

12. An example of 'focus and context' system in a visualisation is
   a) A zoomed-in view of a scatterplot, where a subset of the data takes up the full screen
   b) A fisheye distortion of a map
   c) A timeline view of data, plotting all occurrences of an event against a time axis

13. A paint company plans to display international sales figures, using a colour map based on its 10 most popular paint colours. Each country's area will be coloured using this scheme. A legend at the side of the map will clearly show what sales each colour shows. What is your advice to the company?
   a) Precise analysis may be difficult, as the colours will almost certainly not represent order effectively.
   b) To support more precise analysis, it would be better to choose 10 colours that are visually quite distinct.
   c) The number of colours is a problem, in that 10 is too many for people to easily use in precise analysis.

14. You are working on a visualisation of a global climate data set: an interactive map. So far, there are four choices of data set that you are considering, two potential interaction techniques that you might use, and three choices of colour map. If you now consider an additional data set, how much bigger will the design space be?
   a) 5 times bigger
   b) 4 times bigger
   c) 6 times bigger

15. The total sales of a company's main product in each of the last five years, taken from a company database is shown in a bar chart. A map, using the same database, shows the distribution of the company's shops, with each mark showing the number of staff working there. The company asks you to link these views. What is the best technical step to take first?
   a) There is no data shared by the two views, so advise the company that linking cannot be done
   b) Change both charts, so that they share a selectable list of attributes, so that a user can select the attribute displayed by the bar height and map mark
   c) Change each chart, so that each has its own selectable list of attributes, so that the user can select the attribute displayed in each one

SECTION B

1. Imagine a data set of cars' current locations and speeds, in some urban area, as tracked by a sensor network. Demonstrate the *generalised selection* interaction technique using this data set. Explain what semantic model of the data you would use, and how you would apply generalised selection to it, to improve interaction and analysis (when compared to normal direct selection of data objects).
[10]

2. Imagine a data set with a significant number of objects (e.g., 1,000,000), and a high number of dimensions (e.g., 100). You have one visualization of a dimensional reduction of the data, made using a spring model. You also have several other chart types that you could use, e.g. bar charts, scatterplots, and selectable lists. Describe, in general terms, how and why you would link multiple views of the data.

[10]

3. Your local health service wishes to offer patients access to a database with, for each patient, a medical record. This has attributes such as medical conditions, prescriptions, hospital visits, and measurements (e.g., heart rate), recorded over time. The health service has a simple web app, offering SQL queries, but it has asked you to present an alternative design based on dynamic queries. Outline such a design, and present arguments for it when compared to the SQL app.

[10]